

IGNOU BCS-012: Basic Mathematics
Term-End Examination, June 2023 – Solved Paper
 Time: 3 Hours Maximum Marks: 100

Note. Question Number 1 is compulsory. Attempt any three questions from the remaining questions.

Question 1

1(a)

If

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$$

and $(A + B)^2 = A^2 + B^2$, find a and b .

Solution. Since

$$(A + B)^2 = A^2 + AB + BA + B^2,$$

the given condition implies

$$AB + BA = O.$$

Now

$$AB = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} = \begin{bmatrix} a - 2b & 3 \\ 2a - b & 3 \end{bmatrix},$$

and

$$BA = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} a + 2 & -2a - 1 \\ b - 2 & -2b + 1 \end{bmatrix}.$$

Therefore

$$AB + BA = \begin{bmatrix} 2a - 2b + 2 & 2 - 2a \\ 2a - 2 & 4 - 2b \end{bmatrix} = O.$$

Thus $2 - 2a = 0$ and $4 - 2b = 0$, giving

$$\boxed{a = 1, \quad b = 2.}$$

1(b)

Show that $n(n + 1)(2n + 1)$ is a multiple of 6 for every natural number n .

Solution. We need to prove divisibility by 2 and by 3.

Among n and $n + 1$, one is even. Hence $n(n + 1)(2n + 1)$ is divisible by 2.

For divisibility by 3, consider three cases:

- If $n \equiv 0 \pmod{3}$, then n is divisible by 3.
- If $n \equiv 2 \pmod{3}$, then $n + 1 \equiv 0 \pmod{3}$.
- If $n \equiv 1 \pmod{3}$, then $2n + 1 \equiv 2(1) + 1 = 3 \equiv 0 \pmod{3}$.

So the product is divisible by 3. Since it is divisible by both 2 and 3, it is divisible by 6.

$$\boxed{6 \mid n(n + 1)(2n + 1)}$$

1(c)

If $1, \omega, \omega^2$ are cube roots of unity, show that

$$(2 - \omega)(2 - \omega^2)(2 - \omega^{10})(2 - \omega^{11}) = 49.$$

Solution. For cube roots of unity,

$$\omega^3 = 1, \quad 1 + \omega + \omega^2 = 0.$$

Hence

$$\omega^{10} = \omega^9 \omega = \omega, \quad \omega^{11} = \omega^9 \omega^2 = \omega^2.$$

So

$$(2 - \omega)(2 - \omega^2)(2 - \omega^{10})(2 - \omega^{11}) = \{(2 - \omega)(2 - \omega^2)\}^2.$$

Now

$$(2 - \omega)(2 - \omega^2) = 4 - 2(\omega + \omega^2) + \omega^3 = 4 - 2(-1) + 1 = 7.$$

Therefore

$$(2 - \omega)(2 - \omega^2)(2 - \omega^{10})(2 - \omega^{11}) = 7^2 = 49.$$

1(d)

Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$.

Solution. Let

$$\vec{u} = |\vec{a}|\vec{b} + |\vec{b}|\vec{a}, \quad \vec{v} = |\vec{a}|\vec{b} - |\vec{b}|\vec{a}.$$

Then

$$\vec{u} \cdot \vec{v} = (|\vec{a}|\vec{b} + |\vec{b}|\vec{a}) \cdot (|\vec{a}|\vec{b} - |\vec{b}|\vec{a}).$$

Using the identity $(p + q) \cdot (p - q) = p \cdot p - q \cdot q$,

$$\vec{u} \cdot \vec{v} = |\vec{a}|^2(\vec{b} \cdot \vec{b}) - |\vec{b}|^2(\vec{a} \cdot \vec{a}) = |\vec{a}|^2|\vec{b}|^2 - |\vec{b}|^2|\vec{a}|^2 = 0.$$

Therefore the two vectors are perpendicular.

$$\vec{u} \perp \vec{v}$$

1(e)

Solve the equation $2x^3 - 15x^2 + 37x - 30 = 0$, given that the roots of the equation are in A.P.

Solution. Since the roots are in A.P., let them be $a - d, a, a + d$. For the cubic equation

$$2x^3 - 15x^2 + 37x - 30 = 0,$$

the sum of the roots is

$$\frac{15}{2}.$$

Thus

$$(a - d) + a + (a + d) = 3a = \frac{15}{2} \Rightarrow a = \frac{5}{2}.$$

So one root is $\frac{5}{2}$. Dividing the polynomial by $x - \frac{5}{2}$, or equivalently by $2x - 5$, we get

$$2x^3 - 15x^2 + 37x - 30 = (2x - 5)(x^2 - 5x + 6).$$

Now

$$x^2 - 5x + 6 = (x - 2)(x - 3).$$

Hence

$$x = 2, \quad \frac{5}{2}, \quad 3.$$

1(f)

Evaluate the integral

$$I = \int \frac{x^2}{(x+1)^3} dx.$$

Solution. Put $u = x + 1$. Then $x = u - 1$ and $dx = du$.

$$I = \int \frac{(u-1)^2}{u^3} du = \int \frac{u^2 - 2u + 1}{u^3} du.$$

Thus

$$I = \int \left(\frac{1}{u} - \frac{2}{u^2} + \frac{1}{u^3} \right) du.$$

Integrating term by term,

$$I = \ln |u| + \frac{2}{u} - \frac{1}{2u^2} + C.$$

Putting $u = x + 1$,

$$I = \ln |x + 1| + \frac{2}{x + 1} - \frac{1}{2(x + 1)^2} + C.$$

1(g)

Use first derivative test to find the local maxima and local minima if $f(x) = x^3 - 12x$.

Solution.

$$f(x) = x^3 - 12x, \quad f'(x) = 3x^2 - 12 = 3(x - 2)(x + 2).$$

Critical points are

$$x = -2, \quad x = 2.$$

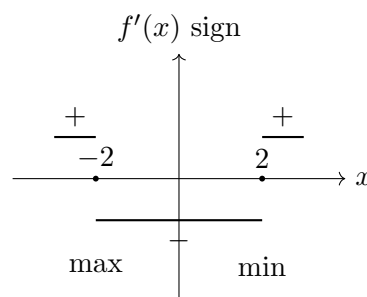
Sign of $f'(x)$:

Interval	$(-\infty, -2)$	$(-2, 2)$	$(2, \infty)$
$f'(x)$	+	-	+

Hence the function changes from increasing to decreasing at $x = -2$, so there is a local maximum. It changes from decreasing to increasing at $x = 2$, so there is a local minimum.

$$f(-2) = (-2)^3 - 12(-2) = 16, \quad f(2) = 8 - 24 = -16.$$

$$\text{Local maximum} = (-2, 16), \quad \text{local minimum} = (2, -16).$$



1(h)

Prove that the three medians of a triangle meet at a point called centroid and divide each of the medians in the ratio 2 : 1.

Solution. Let the vertices of the triangle be

$$A(x_1, y_1), \quad B(x_2, y_2), \quad C(x_3, y_3).$$

The midpoint of BC is

$$D\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}\right).$$

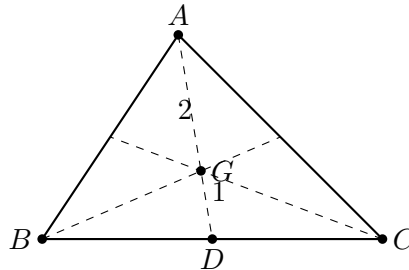
The point which divides median AD in the ratio 2 : 1 from A to D is

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

Similarly, the midpoint of CA and the midpoint of AB give the same point G on the other two medians. Hence all three medians pass through the same point

$$G\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

This common point is called the centroid. Since G divides each median in the ratio 2 : 1 from the vertex, the result is proved.



Question 2

2(a)

Verify that $2 + 2^2 + \dots + 2^n = 2^{n+1} - 2$ using the principle of mathematical induction.

Solution. For $n = 1$,

$$\text{LHS} = 2, \quad \text{RHS} = 2^2 - 2 = 2.$$

So the result is true for $n = 1$.

Assume it is true for $n = k$:

$$2 + 2^2 + \dots + 2^k = 2^{k+1} - 2.$$

For $n = k + 1$,

$$2 + 2^2 + \dots + 2^k + 2^{k+1} = (2^{k+1} - 2) + 2^{k+1} = 2 \cdot 2^{k+1} - 2 = 2^{k+2} - 2.$$

This is the required formula for $k + 1$. Hence, by induction,

$$2 + 2^2 + \dots + 2^n = 2^{n+1} - 2.$$

2(b)

Determine the 10th term of the Harmonic Progression

$$\frac{1}{7}, \frac{1}{15}, \frac{1}{23}, \dots$$

Solution. The reciprocals of an H.P. form an A.P. Thus

$$7, 15, 23, \dots$$

is an A.P. with first term $a = 7$ and common difference $d = 8$. The 10th term of this A.P. is

$$T_{10} = a + 9d = 7 + 9(8) = 79.$$

Therefore, the 10th term of the original H.P. is

$$\boxed{\frac{1}{79}}.$$

2(c)

Evaluate

$$\int \frac{dx}{\sqrt{x} + x}.$$

Solution. Put $t = \sqrt{x}$. Then $x = t^2$ and $dx = 2t dt$.

$$\int \frac{dx}{\sqrt{x} + x} = \int \frac{2t dt}{t + t^2} = \int \frac{2t dt}{t(1 + t)} = \int \frac{2}{1 + t} dt.$$

Therefore

$$\int \frac{dx}{\sqrt{x} + x} = 2 \ln |1 + t| + C.$$

Putting $t = \sqrt{x}$,

$$\boxed{\int \frac{dx}{\sqrt{x} + x} = 2 \ln |1 + \sqrt{x}| + C.}$$

2(d)

Solve the system of equations using Cramer's rule:

$$x + 2y + 2z = 3, \quad 3x - 2y + z = 4, \quad x + y + z = 2.$$

Solution. The coefficient determinant is

$$\Delta = \begin{vmatrix} 1 & 2 & 2 \\ 3 & -2 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 5.$$

Replacing the first, second and third columns by constants respectively,

$$\Delta_x = \begin{vmatrix} 3 & 2 & 2 \\ 4 & -2 & 1 \\ 2 & 1 & 1 \end{vmatrix} = 5,$$

$$\Delta_y = \begin{vmatrix} 1 & 3 & 2 \\ 3 & 4 & 1 \\ 1 & 2 & 1 \end{vmatrix} = 0,$$

$$\Delta_z = \begin{vmatrix} 1 & 2 & 3 \\ 3 & -2 & 4 \\ 1 & 1 & 2 \end{vmatrix} = 5.$$

By Cramer's rule,

$$x = \frac{\Delta_x}{\Delta} = 1, \quad y = \frac{\Delta_y}{\Delta} = 0, \quad z = \frac{\Delta_z}{\Delta} = 1.$$

$$\boxed{x = 1, \quad y = 0, \quad z = 1.}$$

Question 3

3(a)

Given

$$x = a + b, \quad y = a\omega + b\omega^2, \quad z = a\omega^2 + b\omega,$$

verify that $xyz = a^3 + b^3$, where ω is a cube root of unity and $\omega \neq 1$.

Solution. For cube roots of unity,

$$1 + \omega + \omega^2 = 0, \quad \omega^3 = 1.$$

Now

$$yz = (a\omega + b\omega^2)(a\omega^2 + b\omega).$$

Therefore

$$yz = a^2\omega^3 + ab(\omega^2 + \omega^4) + b^2\omega^3.$$

Since $\omega^3 = 1$ and $\omega^4 = \omega$,

$$yz = a^2 + b^2 + ab(\omega + \omega^2) = a^2 + b^2 - ab.$$

Thus

$$xyz = (a + b)(a^2 - ab + b^2) = a^3 + b^3.$$

$$\boxed{xyz = a^3 + b^3.}$$

3(b)

Given

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix},$$

perform the following:

(i) Determine A^{-1} and A^3 .

(ii) Verify that $A^2 - 4A - 5I_3 = O$.

Solution. Using elementary operations or the adjoint method,

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}.$$

Also,

$$A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}.$$

Hence

$$A^2 - 4A - 5I_3 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = O.$$

Thus the identity is verified.

Further,

$$A^3 = A \cdot A^2 = \begin{bmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}.$$

3(c)

If the roots of $ax^3 + bx^2 + cx + d = 0$ are in A.P., show that

$$2b^3 - 9abc + 27a^2d = 0.$$

Solution. Let the roots be

$$r - s, \quad r, \quad r + s.$$

For the cubic $ax^3 + bx^2 + cx + d = 0$,

$$(r - s) + r + (r + s) = 3r = -\frac{b}{a}.$$

Thus

$$r = -\frac{b}{3a}.$$

The product of the roots is

$$(r - s)r(r + s) = r(r^2 - s^2) = -\frac{d}{a}.$$

Also the sum of pairwise products is

$$(r - s)r + r(r + s) + (r - s)(r + s) = 3r^2 - s^2 = \frac{c}{a}.$$

Hence

$$s^2 = 3r^2 - \frac{c}{a}.$$

So

$$r(r^2 - s^2) = r \left(r^2 - 3r^2 + \frac{c}{a} \right) = r \left(\frac{c}{a} - 2r^2 \right).$$

Using $r = -\frac{b}{3a}$,

$$-\frac{d}{a} = -\frac{b}{3a} \left(\frac{c}{a} - \frac{2b^2}{9a^2} \right).$$

Multiplying by $27a^3$,

$$-27a^2d = -9abc + 2b^3.$$

Therefore

$$2b^3 - 9abc + 27a^2d = 0.$$

Question 4**4(a)**

Determine the points of local extrema of

$$f(x) = \frac{3}{4}x^4 - 8x^3 + \frac{45}{2}x^2 + 2015.$$

Solution.

$$f'(x) = 3x^3 - 24x^2 + 45x = 3x(x-3)(x-5).$$

So the critical points are

$$x = 0, \quad 3, \quad 5.$$

Now

$$f''(x) = 9x^2 - 48x + 45.$$

At $x = 0$,

$$f''(0) = 45 > 0,$$

so $x = 0$ gives local minimum.At $x = 3$,

$$f''(3) = 81 - 144 + 45 = -18 < 0,$$

so $x = 3$ gives local maximum.At $x = 5$,

$$f''(5) = 225 - 240 + 45 = 30 > 0,$$

so $x = 5$ gives local minimum.

The function values are

$$f(0) = 2015, \quad f(3) = \frac{8249}{4}, \quad f(5) = \frac{8185}{4}.$$

$$\text{Local minima: } (0, 2015), \left(5, \frac{8185}{4}\right), \quad \text{local maximum: } \left(3, \frac{8249}{4}\right).$$

4(b)

Calculate the shortest distance between the lines

$$\vec{r}_1 = (1 + \lambda)\hat{i} + (2 - \lambda)\hat{j} + (1 + \lambda)\hat{k}$$

and

$$\vec{r}_2 = 2(1 + \mu)\hat{i} + (2 - \mu)\hat{j} + (-1 + 2\mu)\hat{k}.$$

Solution. The first line passes through

$$P_1 = (1, 2, 1)$$

and has direction vector

$$\vec{d}_1 = (1, -1, 1).$$

The second line passes through

$$P_2 = (2, 2, -1)$$

and has direction vector

$$\vec{d}_2 = (2, -1, 2).$$

For two skew lines, the shortest distance is

$$D = \frac{|(\vec{P}_2 - \vec{P}_1) \cdot (\vec{d}_1 \times \vec{d}_2)|}{|\vec{d}_1 \times \vec{d}_2|}.$$

Now

$$\vec{P}_2 - \vec{P}_1 = (1, 0, -2),$$

and

$$\vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & -1 & 2 \end{vmatrix} = (-1, 0, 1).$$

Thus

$$(\vec{P}_2 - \vec{P}_1) \cdot (\vec{d}_1 \times \vec{d}_2) = (1, 0, -2) \cdot (-1, 0, 1) = -3.$$

Also

$$|\vec{d}_1 \times \vec{d}_2| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}.$$

Therefore

$$D = \frac{3}{\sqrt{2}}.$$

4(c)

Determine the values of x for which

$$f(x) = 5x^{3/2} - 3x^{5/2}, \quad x > 0$$

is increasing and decreasing.

Solution.

$$f'(x) = 5 \cdot \frac{3}{2}x^{1/2} - 3 \cdot \frac{5}{2}x^{3/2} = \frac{15}{2}x^{1/2} - \frac{15}{2}x^{3/2}.$$

Thus

$$f'(x) = \frac{15}{2}\sqrt{x}(1 - x).$$

Since $x > 0$, we have $\sqrt{x} > 0$. Therefore the sign of $f'(x)$ depends on $1 - x$.

$$f'(x) > 0 \quad \text{if } 0 < x < 1,$$

and

$$f'(x) < 0 \quad \text{if } x > 1.$$

Therefore

$$\boxed{\text{Increasing on } (0, 1), \quad \text{decreasing on } (1, \infty).}$$

4(d)

If $|z - 2i| = |z + 2i|$, verify that $\text{Im}(z) = 0$.

Solution. Let

$$z = x + iy.$$

Then

$$z - 2i = x + i(y - 2), \quad z + 2i = x + i(y + 2).$$

Given

$$|z - 2i| = |z + 2i|.$$

Squaring both sides,

$$x^2 + (y - 2)^2 = x^2 + (y + 2)^2.$$

Cancelling x^2 ,

$$y^2 - 4y + 4 = y^2 + 4y + 4.$$

Thus

$$-4y = 4y \Rightarrow y = 0.$$

Since $\text{Im}(z) = y$,

$$\boxed{\text{Im}(z) = 0.}$$

Question 5

5(a)

Find the direction cosines of the line passing through the points $(1, 2, 3)$ and $(-1, 1, 0)$.

Solution. The direction ratios of the line are

$$(-1 - 1, 1 - 2, 0 - 3) = (-2, -1, -3).$$

The magnitude is

$$\sqrt{(-2)^2 + (-1)^2 + (-3)^2} = \sqrt{14}.$$

Hence the direction cosines are

$$\boxed{\left(-\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, -\frac{3}{\sqrt{14}}\right)}.$$

If the opposite direction is chosen, all signs become positive. Both represent the same line.

5(b)

Find the area bounded by the curves $y = x^2$ and $y = x$.

Solution. The points of intersection are found by solving

$$x^2 = x.$$

Thus

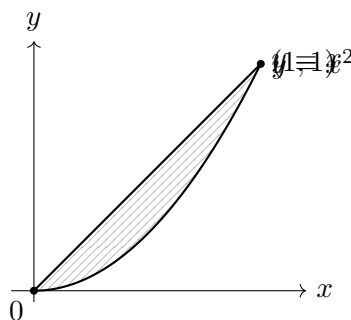
$$x(x - 1) = 0 \Rightarrow x = 0, 1.$$

For $0 \leq x \leq 1$, the upper curve is $y = x$ and the lower curve is $y = x^2$. Therefore the required area is

$$A = \int_0^1 (x - x^2) dx.$$

$$A = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

$$\boxed{\text{Area} = \frac{1}{6} \text{ square units.}}$$



5(c)

Two tailors A and B earn Rs. 150 and Rs. 200 per day respectively. Tailor A can stitch 6 shirts and 4 pants while Tailor B can stitch 10 shirts and 4 pants per day. How many days shall each work if it is desired to produce at least 60 shirts and 32 pants at minimum labour cost? Also calculate the least cost.

Solution. Let tailor A work for x days and tailor B work for y days.

The constraints are

$$6x + 10y \geq 60 \Rightarrow 3x + 5y \geq 30,$$

$$4x + 4y \geq 32 \Rightarrow x + y \geq 8,$$

$$x \geq 0, \quad y \geq 0.$$

The cost function to be minimized is

$$C = 150x + 200y.$$

The corner points of the feasible region relevant to minimum cost are obtained from the boundary lines.

Intersection of

$$3x + 5y = 30, \quad x + y = 8 :$$

from $x = 8 - y$,

$$3(8 - y) + 5y = 30 \Rightarrow 24 + 2y = 30 \Rightarrow y = 3,$$

so

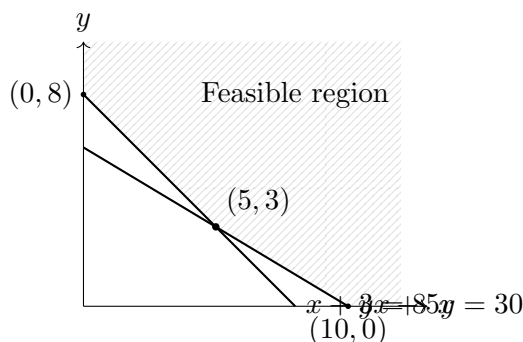
$$x = 5.$$

Cost at $(5, 3)$:

$$C = 150(5) + 200(3) = 750 + 600 = 1350.$$

Other feasible boundary points give higher cost: $(10, 0)$ gives Rs. 1500 and $(0, 8)$ gives Rs. 1600. Therefore the minimum occurs at $(5, 3)$.

Tailor A should work 5 days and Tailor B should work 3 days. Minimum cost = Rs. 1350.



Theory and Formula Sheet

Matrices

- $(A + B)^2 = A^2 + AB + BA + B^2$.
- If $(A + B)^2 = A^2 + B^2$, then $AB + BA = O$.
- Cramer's rule: for $AX = B$, $x_i = \Delta_i / \Delta$.
- $A^{-1} = \frac{1}{|A|} \text{adj}(A)$, if $|A| \neq 0$.

Progressions

- A.P.: $T_n = a + (n - 1)d$.
- H.P.: reciprocals form an A.P.
- G.P. finite sum: $S_n = a \frac{r^n - 1}{r - 1}$ for $r \neq 1$.

Cube Roots of Unity

- $\omega^3 = 1, \omega \neq 1$.
- $1 + \omega + \omega^2 = 0$.
- Powers repeat cyclically: $\omega^{3k} = 1, \omega^{3k+1} = \omega, \omega^{3k+2} = \omega^2$.

Complex Numbers

- If $z = x + iy$, then $|z| = \sqrt{x^2 + y^2}$.
- $\text{Re}(z) = x, \text{Im}(z) = y$.
- Equate squared moduli to remove square roots.

Differentiation and Extrema

- Critical points satisfy $f'(x) = 0$ or $f'(x)$ undefined.

- First derivative test: $+$ to $-$ gives local maximum; $-$ to $+$ gives local minimum.
- Second derivative test: $f''(a) > 0$ minimum; $f''(a) < 0$ maximum.

Integration

- Use substitution when radicals or shifted powers occur.
- $\int \frac{du}{u} = \ln |u| + C$.
- $\int u^n du = \frac{u^{n+1}}{n+1} + C, n \neq -1$.

Vectors and 3D Geometry

- Perpendicular vectors have zero dot product.
- Direction cosines: $l = \frac{a}{\sqrt{a^2+b^2+c^2}}, m = \frac{b}{\sqrt{a^2+b^2+c^2}}, n = \frac{c}{\sqrt{a^2+b^2+c^2}}$.
- Shortest distance between skew lines:

$$D = \frac{|(\vec{P}_2 - \vec{P}_1) \cdot (\vec{d}_1 \times \vec{d}_2)|}{|\vec{d}_1 \times \vec{d}_2|}$$

Area and Linear Programming

- Area between curves: $\int_a^b (\text{upper} - \text{lower}) dx$.
- In linear programming, optimum occurs at a corner point of the feasible region.
- For minimization, evaluate the cost function at all relevant corner points.